

Team Coding Work Distribution

Absar, Harshit and Maham: three equal workstreams, frozen interface contracts and a conflict-free merge plan

Machine Learning for Alpha Signal Combination and Portfolio Construction

All three members are equal partners with the same planned workload (100 points each). Work proceeds in parallel against synthetic data and jointly frozen contracts, with rotating integration checkpoints, so the final merge is a formality rather than a risk.

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Team Coding Work Distribution

1. Purpose and Principles

This document splits the coding work for the selected paper (*Where Does Machine-Learning Alpha Come From After Trading Costs?*) among three team members. **All three are equal partners.** Nobody is a lead, nobody holds extra approval rights, and every member carries the same planned workload. Work runs **in parallel** and merges at the end **without conflicts**. Tasks map to the experiment IDs (E00-E64) in Workbook 4 and the specification in PDF 4. Members are listed alphabetically throughout.

Team and workstreams

Member	Background	Workstream	Planned effort
Absar	Data engineering	Workstream A: data, transaction costs and backtesting	100 points (1/3)
Harshit	ML modelling; portfolio construction; data engineering	Workstream B: models and portfolio construction	100 points (1/3)
Maham	Statistics & reporting	Workstream C: baselines, statistics and reporting	100 points (1/3)

How the split was balanced

- **Equal effort.** Every task has an effort estimate in planning points (Section 5), and each member's tasks sum to exactly 100. Points are planning estimates. If actual effort in one workstream drifts more than 15% from the others, tasks are rebalanced at the next checkpoint by agreement of all three.
- **Matched to background.** Each workstream fits the member's stated strengths. No workstream is treated as more central than another: the paper needs all three to produce a single result.
- **Shared duties are shared.** The contracts, split calendar, integration script, specification freeze and lockbox run belong to the whole team. The duty of running each integration checkpoint rotates (Section 6).
- **Symmetric review.** Pull requests are reviewed in a circle (Absar reviews Harshit, Harshit reviews Maham, Maham reviews Absar). Changes to shared files need all three approvals.

Five rules that make parallel work mergeable

1. **Contracts first.** In Week 1 all three members jointly agree and freeze every data schema, file path and function signature that crosses a workstream boundary (Section 4). Afterwards, contracts change only through a Contract Change Request approved by all three.
2. **Folder ownership.** Each member edits only their workstream's folders (Section 3). Nobody edits another member's files, so Git merge conflicts cannot occur in source code.
3. **Synthetic data from day one.** A synthetic dataset with the exact real schema is available in Week 1. Everyone builds and tests against it, so nobody waits for anybody else.
4. **Run-ID keyed outputs.** Every output file is written under its own `run_id`, so runs never overwrite each other.
5. **Continuous contract tests.** CI validates every output against the frozen schemas on every pull request. Integration problems surface in days, not at the end.

2. Overview Diagrams

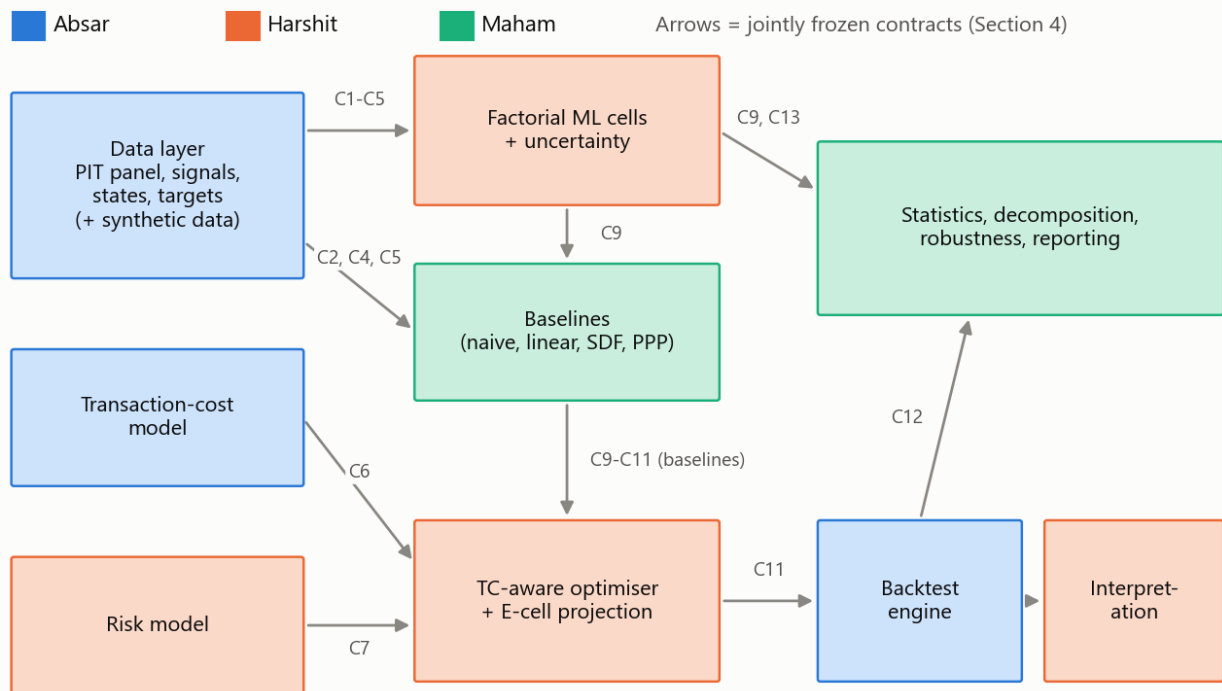


Figure 1. Workstream ownership and the artefacts (contracts) that flow between members. Every arrow is a frozen schema validated in CI.

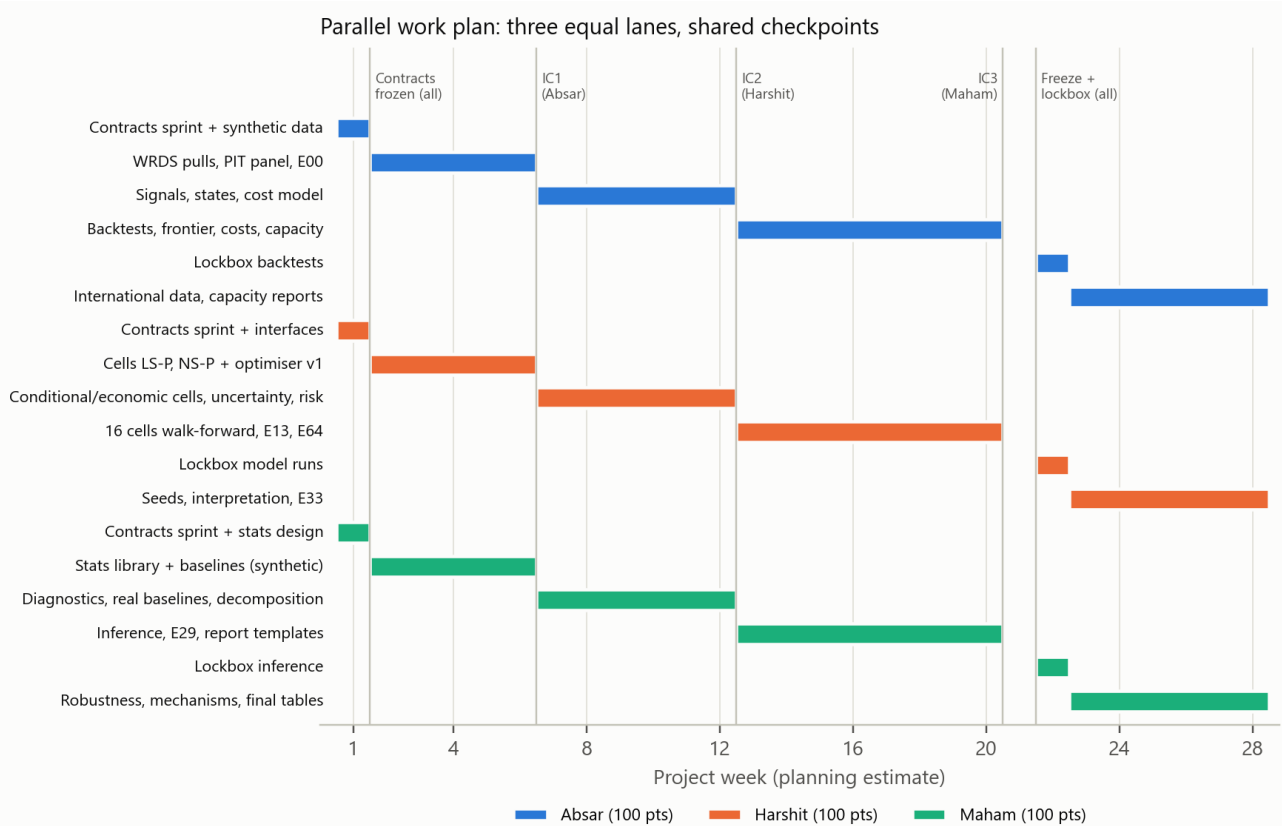


Figure 2. Parallel work plan by week (planning estimate). Vertical lines mark integration checkpoints; all three lanes run simultaneously.

3. Repository Layout and Ownership (CODEOWNERS)

```

alphacomb/
  pyproject.toml, environment.yml, .pre-commit-config.yaml  SHARED (frozen Week 1; changes need 3 approvals)
  configs/base.yaml, configs/splits.yaml  SHARED
  configs/data.yaml, configs/costs.yaml, configs/backtest.yaml  ABSAR (A)
  configs/models.yaml, configs/portfolio.yaml, configs/risk.yaml  HARSHIT (B)
  configs/stats.yaml, configs/robustness/*.yaml  MAHAM (C)
  src/alphacomb/contracts/  schemas.py, io.py, paths.py, splits.py, VERSION  SHARED
  src/alphacomb/synthetic/  synthetic data generator (same schemas)  ABSAR (A)
  src/alphacomb/data/  WRDS pulls, PIT panel, universe, signals, states  ABSAR (A)
  src/alphacomb/costs/  spreads, impact function, borrow fees  ABSAR (A)
  src/alphacomb/backtest/  P&L engine, execution lag, drift, cost ledger  ABSAR (A)
  src/alphacomb/models/  16 factorial cells, uncertainty, published ML  HARSHIT (B)
  src/alphacomb/portfolio/  TC-aware optimiser, constraints, E-cell projection  HARSHIT (B)
  src/alphacomb/risk/  structural factor risk model  HARSHIT (B)
  src/alphacomb/interpret/  SHAP/ALE, economic feature importance  HARSHIT (B)
  src/alphacomb/baselines/  naive, linear, KNS SDF, linear PPP, banding  MAHAM (C)
  src/alphacomb/stats/  NW, LW2008, CW/DM, SPA, StepM, MCS, DSR, PBO, decomposition  MAHAM (C)
  src/alphacomb/reporting/  tables, figures, LaTeX export  MAHAM (C)
  pipelines/01_build_data.py, 05_backtest.py  ABSAR (A)
  pipelines/02_train_models.py, 04_construct_portfolios.py  HARSHIT (B)
  pipelines/03_baselines.py, 06_inference.py, 07_report.py  MAHAM (C)
  pipelines/run_all.py  SHARED (only calls stage scripts; no logic)
  tests/contracts/  SHARED
  tests/data, tests/costs, tests/backtest, tests/synthetic  ABSAR (A)
  tests/models, tests/portfolio, tests/risk, tests/interpret  HARSHIT (B)
  tests/baselines, tests/stats, tests/reporting  MAHAM (C)
  data/ outputs/ mlrns/  git-ignored (never committed)

```

4. Interface Contracts (agreed jointly and frozen in Week 1)

All tables are Parquet, keyed by date (month-end, `datetime64[ns]`) and `permno (int32)`. Schemas are enforced with `pandera` in `src/alphacomb/contracts/schemas.py`. Paths are generated only by `contracts/paths.py`; nobody hard-codes file paths.

#	Artefact (path)	Produced by	Used by	Required columns	First version
C1	<code>data/processed/universe.parquet</code>	Absar	All	<code>date</code> , <code>permno</code> , <code>in_universe (bool)</code> , <code>me</code> , <code>price</code> , <code>exchcd</code> , <code>ff49</code> , <code>nyse_size_pct</code>	Synthetic W1; real W9
C2	<code>data/processed/signals.parquet</code>	Absar	Harshit, Maham	<code>date</code> , <code>permno</code> , <code>sig_* (float32 in [-0.5, 0.5])</code> , <code>miss_<theme> (int8)</code>	Synthetic W1; real W9
C3	<code>data/processed/signal_meta.csv</code>	Absar	Harshit, Maham	<code>signal</code> , <code>theme</code> , <code>pub_year</code> , <code>source</code>	W1
C4	<code>data/processed/targets.parquet</code>	Absar	Harshit, Maham	<code>date</code> , <code>permno</code> , <code>r_1m</code> , <code>r_3m</code> , <code>r_6m</code> , <code>r_12m</code> (<code>excess</code> , <code>delisting-adjusted</code>), <code>ret_next</code>	Synthetic W1; real W9
C5	<code>data/processed/states.parquet</code>	Absar	Harshit, Maham	<code>date</code> , <code>MKTVOL</code> , <code>BEAR</code> , <code>ILLIQ</code> , <code>SENT</code> , <code>CREDIT</code> , <code>TERM</code> , <code>INFL</code> , <code>DRATE</code> , <code>FMOM_<theme></code> , <code>DISP</code>	Synthetic W1; real W10
C6	<code>data/processed/cost_inputs.parquet</code>	Absar	Harshit, Maham	<code>date</code> , <code>permno</code> , <code>spread</code> , <code>sigma_d</code> , <code>adv_usd</code> , <code>borrow_fee</code>	Synthetic W1; real W12
C7	<code>risk.load(date) -> RiskModel(B, F, D)</code>	Harshit	Harshit, Maham	B: <code>permno × factor DataFrame</code> ; F: <code>factor covariance</code> ; D: <code>specific variance Series</code>	Synthetic W2; real W12
C8	<code>configs/splits.yaml + contracts/splits.py</code>	All (Week 1 sprint)	All	<code>train/validation/test/lockbox calendar</code> , <code>embargo rule</code>	W1
C9	<code>outputs/predictions/{strategy}/{run_id}.parquet</code>	Harshit (cells), Maham (baselines)	Harshit, Maham	<code>date</code> , <code>permno</code> , <code>score</code> , <code>alpha</code> , <code>unc_sd (nullable)</code>	Synthetic W5
C10	<code>outputs/weight_proposals/{strategy}/{run_id}.parquet</code>	Harshit (E-cells), Maham (linear PPP)	Harshit	<code>date</code> , <code>permno</code> , <code>w_prop</code>	Synthetic W8
C11	<code>outputs/weights/{strategy}/{run_id}.parquet</code>	Harshit (optimiser), Maham (banding)	Absar, Harshit, Maham	<code>date</code> , <code>permno</code> , <code>w</code>	Synthetic W6

#	Artefact (path)	Produced by	Used by	Required columns	First version
C12	outputs/returns/{strategy}/{run_id}.parquet	Absar (backtest)	Harshit, Maham	date, gross_ret, net_ret, turnover, cost_spread, cost_impact, cost_borrow, long_ret, short_ret	Synthetic W6
C13	outputs/trials.csv (append-only via contracts/io.log_trial)	Everyone who trains or tunes	Maham (DSR, PBO, SPA)	run_id, strategy, cell, params_json, seed, train_end, val_metric, git_sha, timestamp	W1

Function signatures (frozen).

```
# contracts/interfaces.py (SHARED, frozen Week 1)
def fit_predict(cell: CellSpec, split: Split, data: DataBundle,
               cfg: dict) -> pd.DataFrame: ... # Harshit/Maham -> C9 rows
def construct(date, alpha: pd.Series, w_prev: pd.Series, risk: RiskModel,
             costs: pd.DataFrame, cfg: dict) -> pd.Series: ... # Harshit -> C11
def trade_cost(dq_usd: pd.Series, cost_row: pd.DataFrame,
              cfg: dict) -> pd.Series: ... # Absar; used by all
def backtest(weights: pd.DataFrame, targets: pd.DataFrame,
             costs: pd.DataFrame, cfg: dict) -> pd.DataFrame: ... # Absar -> C12
def run_tests(returns: dict[str, pd.DataFrame], trials: pd.DataFrame,
             cfg: dict) -> dict[str, pd.DataFrame]: ... # Maham -> tables
```

Contract Change Request (CCR). A CCR is a pull request that edits only contracts/ or tests/contracts/. It bumps contracts/VERSION, describes the migration, and needs approval from all three members. Each member then updates their own modules in separate PRs. Additive changes, such as a new nullable column, are preferred over renames.

5. Work Assignment by Member (equal effort: 100 points each)

Planning points are relative effort estimates, not hours or importance. Every member also carries the same 10 points of shared duties: the Week 1 contracts sprint (4), one rotating integration checkpoint (3) and peer review (3).

5.1 Absar: Workstream A (data, transaction costs, backtesting)

Area	Tasks	Exp. IDs	Deliverables	Acceptance criteria	Points
Synthetic data	Generator producing C1-C6 with realistic shapes (e.g., 500 stocks × 300 months), planted effects, missingness, delistings, cost ranges	-	synthetic/generate.py, data/synthetic/*	Passes all contract tests; used by everyone from Week 2	10
WRDS extraction & PIT panel	CRSP monthly/daily, delistings, Compustat + CCM, accounting lags, universe filters	E00	data/wrds_pull.py, data/pit_panel.py, C1, C4	Rebuilt FF factors correlate > 0.95 with Ken French; zero look-ahead test failures	18
Signal library	~150 JKP/OpenAP signals, rank normalisation, missing indicators, publication years, gated-library flags	E01	data/signals/*, C2, C3	Replication slope/R ² vs Chen-Zimmermann reported	16
State variables	MKTVOL, BEAR, ILLIQ, SENT, CREDIT, TERM, INFL, DRATE, FMOM, DISP; placebo versions	-	data/states.py, C5, states_placebo.parquet	No future information (tests); placebo files for E53	6
Transaction-cost model	Spread hierarchy (TAQ/EDGE/Abdi-Rinaldo), ADV, volatility, borrow fees; trade_cost()	E31	costs/*, C6	Spread-estimator correlation with TAQ reported; 0.5×-3× multipliers configurable	14
Backtest engine	Execution lag, drifted weights, gross/net P&L, cost ledger, AUM scaling, reconciliation; frontier and capacity runs	E30, E32	backtest/engine.py, C12	Contributions reconcile to portfolio return (tolerance 1e-8); AUM grid and capacity outputs	20
International data	JKP developed ex-US panel in the same schemas	supports E55	data/international/*	Contract-valid files	6
Shared duties	Contracts sprint; runs checkpoint IC1 ; peer review of Harshit's PRs	-	IC1 report	IC1 passes	10
Total					100

5.2 Harshit: Workstream B (models, portfolio construction)

Area	Tasks	Exp. IDs	Deliverables	Acceptance criteria	Points
Factorial ML cells	16 cells: linear/nonlinear × static/state × prediction/economic loss; gated mixture-of-experts; economic-loss NN policy	E20-E27	models/cells/*, confs/models.yaml	Every cell runs on synthetic (IC1) and real data (IC3); all trials logged to C13	28
Uncertainty	Deep ensembles and bagging, κ -shrinkage, bootstrap-SE variant	E28	models/uncertainty/*	unc_sd populated; κ chosen on validation only	7
Published ML benchmarks	GKX-style NN3 and LightGBM; JKMP Portfolio-ML replication from authors' code	E13	models/benchmarks/* producing C9/C10	Directional replication documented	9
Portfolio optimiser	CVXPY TC-aware mean-variance; neutralities, position/trade/ADV limits; E-cell projection; long-only variant	E33	portfolio/*, configs/portfolio.yaml	Constraint violations < 0.1% of months; solver failures logged	18
Risk model	Structural factor model (beta, 13 themes, FF49); EWMA factor/specific risk; PCA alternative; alignment diagnostics	E64	risk/*, C7	Bias statistics reported; risk.load(date) < 2 s per month	14
Seeds & stability	10+ seeds; ± 1 grid-step perturbations	E56	Seed-dispersion outputs	Dispersion table shared with the team	5
Interpretation	Economic feature importance (leave-theme-out), TreeSHAP/deep SHAP, ALE	E60	interpret/*, figure-ready CSVs	Results tied to hypothesis H3	9
Shared duties	Contracts sprint; runs checkpoint IC2 ; peer review of Maham's PRs	-	IC2 report	IC2 passes	10
Total					100

5.3 Maham: Workstream C (baselines, statistics, reporting)

Area	Tasks	Exp. IDs	Deliverables	Acceptance criteria	Points
Statistics library	Newey-West, Ledoit-Wolf Sharpe tests, Clark-West, Diebold-Mariano, Hansen SPA, White RC, Romano-Wolf StepM, MCS, DSR, PBO, Harvey-Liu haircut, BH-FDR, Bai-Perron	-	stats/* with unit tests on simulated series	Known size/power reproduced on simulations	17
Signal diagnostics	IC, rank IC, ICIR, IC half-life, turnover; redundancy clustering	E02, E03	baselines/diagnostics.py, tables	Synthetic by W6, real by W11	7
Baselines	EW/theme-EW/IC/ICIR/inverse-vol composites; Lewellen FM, ridge, lasso, ENet, PLS; KNS shrunk SDF; linear PPP with costs (DMNU); NMV buy/hold banding	E10, E11, E12, E14, E34	baselines/* producing C9/C10/C11	Same tuning budget as model cells; contract tests pass	17
Factorial decomposition	Orthogonal-contrast regressions, Shapley attribution, block-bootstrap CIs	E29	stats/decomposition.py	Recovers planted effects on synthetic data	8
Inference	SPA/StepM/MCS, DSR/PBO over C13, factor alphas, multiverse non-standard errors	E40-E43	pipelines/06_inference.py, tables	Every pre-registered test reported	14
Robustness orchestration	Config grids for subperiods, universes, horizons, placebo states, gated library, international	E50-E55	configs/robustness/*.yaml, appendix tables	All runs reproducible from configs alone	11
Mechanisms & implied weights	Implied state-dependent weights; limits-to-arbitrage tilts; trading speed vs liquidity state	E61, E62, E63	Regression tables	Linked to hypotheses H5/H6	7
Reporting	Publication tables and figures, LaTeX export, experiment-status dashboard	-	reporting/*, pipelines/07_report.py	One command regenerates all paper tables	9
Shared duties	Contracts sprint; runs checkpoint IC3 ; peer review of Absar's PRs	-	IC3 report	IC3 passes	10
Total					100

Workload check

Member	Own tasks	Shared duties	Total	Share
Absar	90	10	100	33.3%
Harshit	90	10	100	33.3%
Maham	90	10	100	33.3%

6. Parallel Timeline and Integration Checkpoints (weeks, planning estimate)

Weeks	Absar	Harshit	Maham	Checkpoint (run by)
1	Contracts sprint (all together); synthetic data generator	Contracts sprint (all together); model and optimiser interfaces	Contracts sprint (all together); stats-library design	Contracts v1.0 frozen by all three
2-6	WRDS pulls; PIT panel; universe; E00	Cells LS-P, NS-P on synthetic; optimiser v1	Stats library + tests; naive/linear baselines on synthetic	IC1 (W6): full synthetic end-to-end (run by Absar)
7-12	Signals E01, states, cost model; real data v1 (W9) and v2 (W12)	Conditional cells, MoE, economic-loss learners, uncertainty; risk model	Diagnostics E02/E03; baselines and E14 on real data; decomposition code	IC2 (W12): real-data end-to-end (run by Harshit)
13-20	Backtest runs; frontier E30; cost sensitivity E31; capacity E32	All 16 cells walk-forward; benchmarks E13; alignment E64	Inference E40-E42; decomposition E29; report templates	IC3 (W20): complete development results (run by Maham)
21	Specification freeze (all three sign)	Specification freeze (all three sign)	Specification freeze (all three sign)	Freeze tag v-freeze (joint)
22	Lockbox backtests	Lockbox model and portfolio runs	Lockbox inference	Lockbox opened once, together
23-28	International data; capacity reports	Seeds E56; interpretation E60; optimiser variants E33	Robustness E50-E55; multiverse E43; mechanisms E61-E63; final tables	Final merge & release tag v-paper (joint)

Nobody is blocked. Before real data exists, everyone uses the synthetic C1-C6. When real data arrives, the only change is a flag in `configs/base.yaml` (`data_source: synthetic | real`).

7. Git Workflow (conflict-free merging)

- Branches.** `main` is protected. Work branches are `absar/<topic>`, `harshit/<topic>` and `maham/<topic>`, short-lived (≤ 1 week) and rebased on `main` daily.
- Pull requests.** A PR touches only its author's owned paths; CI rejects PRs that modify another member's files (CODEOWNERS plus a path check). Review is circular and equal: Absar reviews Harshit, Harshit reviews Maham, Maham reviews Absar. Shared files need all three approvals.
- CI on every PR.** Formatting (ruff/black via pinned pre-commit), unit tests, **contract tests on synthetic data**, and a 5-minute smoke run of the whole pipeline on a tiny synthetic sample.
- No shared mutable state.** Outputs are written under `run_id`. The trial log is append-only through `contracts/io.log_trial` with a file lock. Nothing in `data/` or `outputs/` is committed.
- Dependencies.** `environment.yaml` is pinned. Adding a package is a small shared PR approved by all three.
- Configs over code.** Robustness variants, cost multipliers and AUM levels are configuration files, so grids run over any workstream's code without editing it.
- Reproducibility.** Every run records `git_sha`, config hash, seed and data version. Checkpoints create tags (`ic1`, `ic2`, `ic3`, `v-freeze`, `v-paper`).
- Weekly 30-minute sync.** Facilitation rotates weekly among the three members. Review the contract-test dashboard, blockers and effort points; any contract concern becomes a CCR within 48 hours.

8. Definition of Done (same for every member)

- Code sits only in the owner's folders, with docstrings and type hints.
- Unit tests pass, and contract tests pass on synthetic data (and on real data after IC2).
- Behaviour is driven by a config file with no hard-coded paths, dates or parameters.

- Every training or tuning run is logged to `outputs/trials.csv`.
- Outputs regenerate from a single pipeline command.
- The PR is reviewed by the circular reviewer, and the experiment status and spent effort points are updated in Workbook 4.

9. Risks to Parallel Work and Mitigations

Risk	Impact	Mitigation	Handled by
Workloads drift apart	Unequal effort	Effort points tracked weekly; rebalance at checkpoints if any workstream differs by more than 15%	All
Synthetic data too unrealistic	Code breaks on real data	Realistic missingness, delistings, fat tails and cost ranges; 2-week buffer before IC2	Absar
Contract changes mid-project	Rework	Additive-only changes; CCR with 3 approvals; version bump	All
WRDS/TAQ access delays	Real data late	Synthetic pipeline continues; OpenAP/JKP public files as interim real signals	Absar
Model compute bottleneck	IC3 slips	LightGBM cells first; NN cells on GPU; reduced development grid	Harshit
Inference bugs found late	Wrong conclusions	Statistics library validated on simulations before real results exist	Maham
Hidden test-period tuning	Invalid lockbox	Split calendar enforced in shared code; lockbox dates blocked until v-freeze exists	All
One member unavailable	Delays	Circular reviewers know neighbouring modules; docs and tests allow hand-over	All